THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · NCERT-TYPE MCQ TEST SET 2

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. NCERT-type: questions in the style of the textbook examples and exercises.

MULTIPLE CHOICE QUESTIONS

50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Sperms are produced in the: 1

(a) ovaries **(b)** testes **(c)** uterus **(d)** oviduct

2. The continuous movement of water between the oceans, atmosphere and land is the: 1

(a) nitrogen cycle **(b)** water (hydrological) cycle
(c) carbon cycle **(d)** rock cycle

3. One breath is about 0.5 litre and a person takes about 20000 breaths a day. The air breathed in a day is about: 1

(a) 100000 litres **(b)** 10000 litres **(c)** 1000 litres **(d)** 100 litres

4. Which of the following is a colloid? 1

(a) air **(b)** fog **(c)** salt in water **(d)** sugar in water

5. The atomicity of sulphur (S_8) is: 1

(a) 1 **(b)** 4 **(c)** 2 **(d)** 8

6. The mass number of an atom equals the number of: 1

(a) protons plus electrons **(b)** protons plus neutrons
(c) neutrons plus electrons **(d)** protons only

7. The area under a velocity-time graph gives: 1

(a) acceleration **(b)** speed **(c)** time **(d)** displacement

8. Action and reaction do not cancel because they: 1

(a) are in the same direction **(b)** act at different times
(c) act on different bodies **(d)** are unequal

9. The law of conservation of energy states that energy: 1

(a) can neither be created nor destroyed but only changed from one form to another
(b) is always lost
(c) can be created
(d) can be destroyed

10. The pitch of a sound depends on its: 1

(a) amplitude **(b)** speed **(c)** frequency **(d)** medium

11. Which of the following is a prokaryote? 1

(a) yeast **(b)** bacterium **(c)** amoeba **(d)** onion cell

12. The tissue responsible for the growth of plants is the: 1

(a) permanent tissue **(b)** meristematic tissue

(c) nervous tissue **(d)** epithelial tissue

13. The science of classification of organisms is: 1

(a) taxonomy **(b)** ecology **(c)** physiology **(d)** genetics

14. Amoeba reproduces by: 1

(a) budding **(b)** binary fission **(c)** spore formation **(d)** fragmentation

15. The solid Earth including rocks, soil and the interior is the: 1

(a) geosphere **(b)** atmosphere **(c)** biosphere **(d)** hydrosphere

16. The conservation of energy is an example of: 1

(a) a unit

(b) a general principle that applies in every process

(c) an estimate

(d) a symbol

17. In chromatography the component which is more soluble in the solvent: 1

(a) moves farther up the paper **(b)** moves down

(c) stays at the base **(d)** disappears

18. The chemical formula of sodium chloride is: 1

(a) $NaCl$ **(b)** $NaCl_2$ **(c)** Na_2Cl **(d)** Na_2Cl_2

19. An atom with a complete outermost shell is generally: 1

(a) positive **(b)** very reactive

(c) radioactive **(d)** chemically unreactive

20. Acceleration due to gravity near the Earth's surface is about: 1

(a) 100 m/s squared **(b)** 1 m/s squared **(c)** 10 m/s squared **(d)** 10 m/s

21. Airbags in cars protect passengers by: 1

(a) stopping the car

(b) reducing the mass

(c) increasing the stopping time and reducing the force

(d) reducing the speed

22. Mechanical advantage = 1

(a) load x effort **(b)** load - effort **(c)** effort / load **(d)** load / effort

23. Bats find their prey in the dark by: 1

(a) light **(b)** smell

(c) infrasonic sounds **(d)** echolocation using ultrasonic waves

24. The smooth ER is concerned with: 1

(a) lipid synthesis and detoxification **(b)** respiration

(c) protein synthesis **(d)** photosynthesis

25. Water and minerals are conducted upward in plants by the: 1

(a) phloem **(b)** collenchyma **(c)** parenchyma **(d)** xylem

26. The 'amphibians of the plant kingdom' are the: 1

(a) pteridophytes **(b)** algae **(c)** bryophytes **(d)** gymnosperms

27. The female reproductive part of a flower is the: 1

(a) petal **(b)** carpel (pistil) **(c)** stamen **(d)** sepal

28. The surface that reflects the least solar radiation is: 1

(a) ocean water or forests **(b)** clouds

(c) ice **(d)** snow

29. A scientific law usually: 1

(a) explains why the pattern exists in every detail

(b) cannot be tested

(c) is a personal opinion

(d) describes a regular pattern observed in nature

30. Which of the following is a homogeneous mixture? 1

(a) muddy water **(b)** brass **(c)** oil and water **(d)** sand and sugar

31. According to the law of conservation of mass, in a chemical reaction the total mass of the reactants: 1

(a) is zero **(b)** is less than the products

(c) equals the total mass of the products **(d)** is greater than the products

32. The nucleus of an atom was discovered by: 1

(a) Bohr **(b)** Dalton **(c)** Thomson **(d)** Rutherford

33. Displacement is a: 1

(a) unit **(b)** vector quantity **(c)** constant **(d)** scalar quantity

34. The SI unit of force is: 1

(a) newton **(b)** pascal **(c)** joule **(d)** watt

35. A girl carrying a box walks horizontally. The work done by the force holding the box is: 1

(a) maximum **(b)** positive **(c)** negative **(d)** zero

36. Sound waves in air are: 1

(a) light waves **(b)** transverse waves

(c) longitudinal waves **(d)** electromagnetic waves

37. Virchow added to the cell theory that: 1

(a) cells are made of atoms **(b)** all cells have walls

(c) cells are visible to the eye **(d)** new cells arise from pre-existing cells

38. The epithelium lining the blood vessels and alveoli is: 1

(a) cuboidal epithelium **(b)** columnar epithelium

(c) ciliated epithelium **(d)** squamous epithelium

39. Hydra and jellyfish belong to: 1

(a) Cnidaria (Coelenterata) **(b)** Porifera

(c) Annelida **(d)** Arthropoda

40. Fusion of the male and female gametes is called: 1

(a) budding **(b)** germination **(c)** pollination **(d)** fertilisation

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41. The ozone layer protects us from: 1

(a) sound **(b)** visible light
(c) infrared radiation **(d)** harmful ultraviolet radiation

42. The gap between a prediction and observation is an opportunity to: 1

(a) change the data **(b)** stop experimenting
(c) blame the instrument **(d)** refine the model

43. Colloidal particles pass through ordinary filter paper because they are: 1

(a) larger than the pores of the paper **(b)** smaller than the pores of the paper
(c) heavier than water **(d)** magnetic

44. The molecular mass of H_2O is (H = 1, O = 16): 1

(a) 18 u **(b)** 17 u **(c)** 16 u **(d)** 20 u

45. An atom has 11 protons and 12 neutrons. Its mass number is: 1

(a) 23 **(b)** 12 **(c)** 34 **(d)** 11

46. A straight line parallel to the time axis on a distance-time graph indicates that the body is: 1

(a) at rest **(b)** accelerating
(c) falling **(d)** moving with uniform speed

47. A rocket moves upward because: 1

(a) friction
(b) the exhaust gases pushed downward exert an equal and opposite force on it
(c) air pushes it
(d) the Earth attracts it

48. Power is the: 1

(a) work done **(b)** rate of doing work
(c) energy **(d)** force

49. The quality or timbre of a sound enables us to: 1

(a) distinguish two sounds of the same pitch and loudness
(b) find its loudness
(c) find its wavelength
(d) measure its speed

50. The organelle called the 'powerhouse of the cell' is the: 1

(a) mitochondrion **(b)** Golgi apparatus **(c)** lysosome **(d)** chloroplast

END OF QUESTION PAPER **Total: 50 Marks**

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